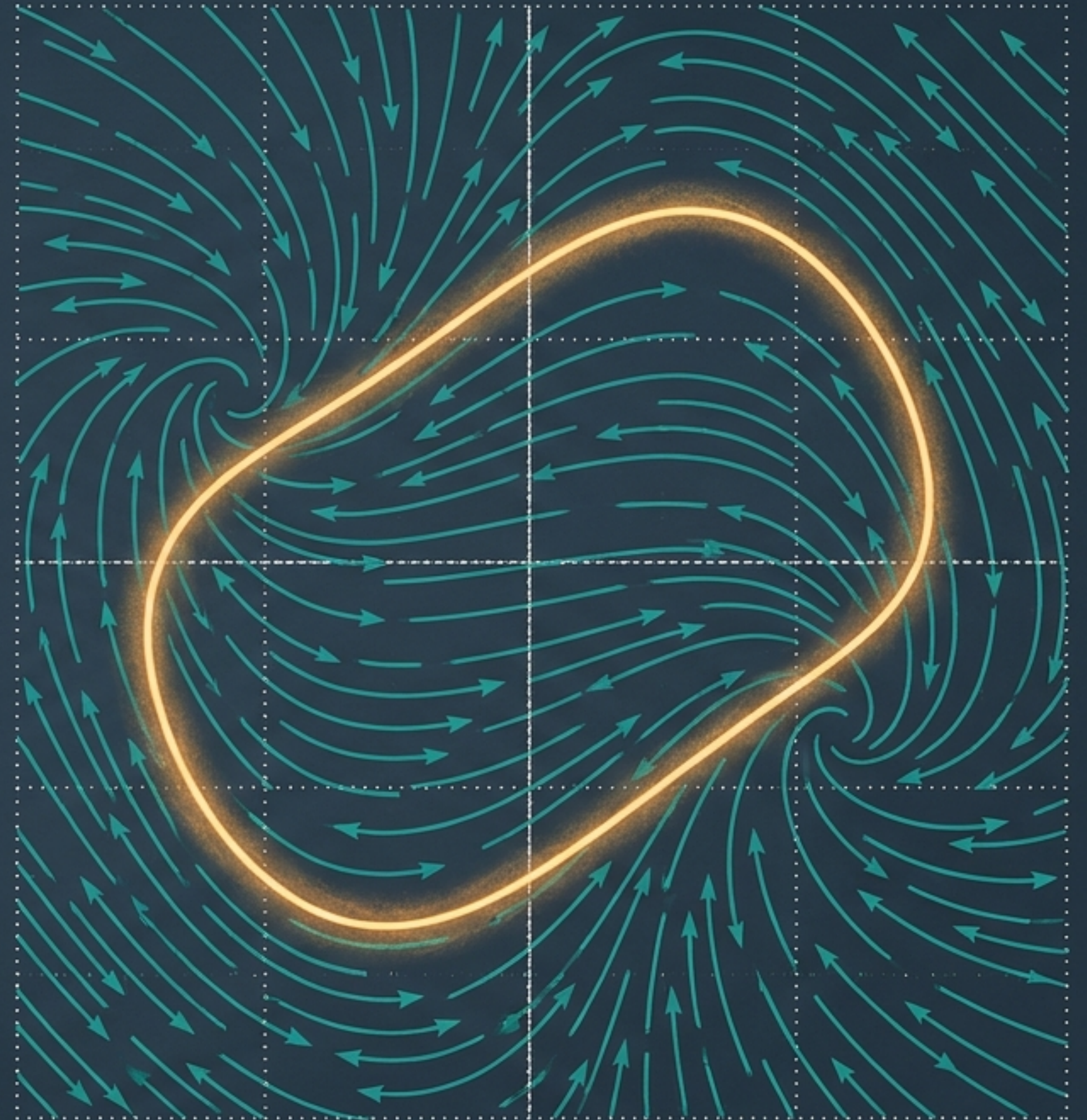


Bounding the Infinite

A Resolution to Hilbert's 16th Problem

Uniform Bounds for Planar Polynomial Limit Cycles

Distilling the 2026 mathematical breakthrough, the 160-page 6-step proof, and the Lean-verified Quintic Liénard companion.

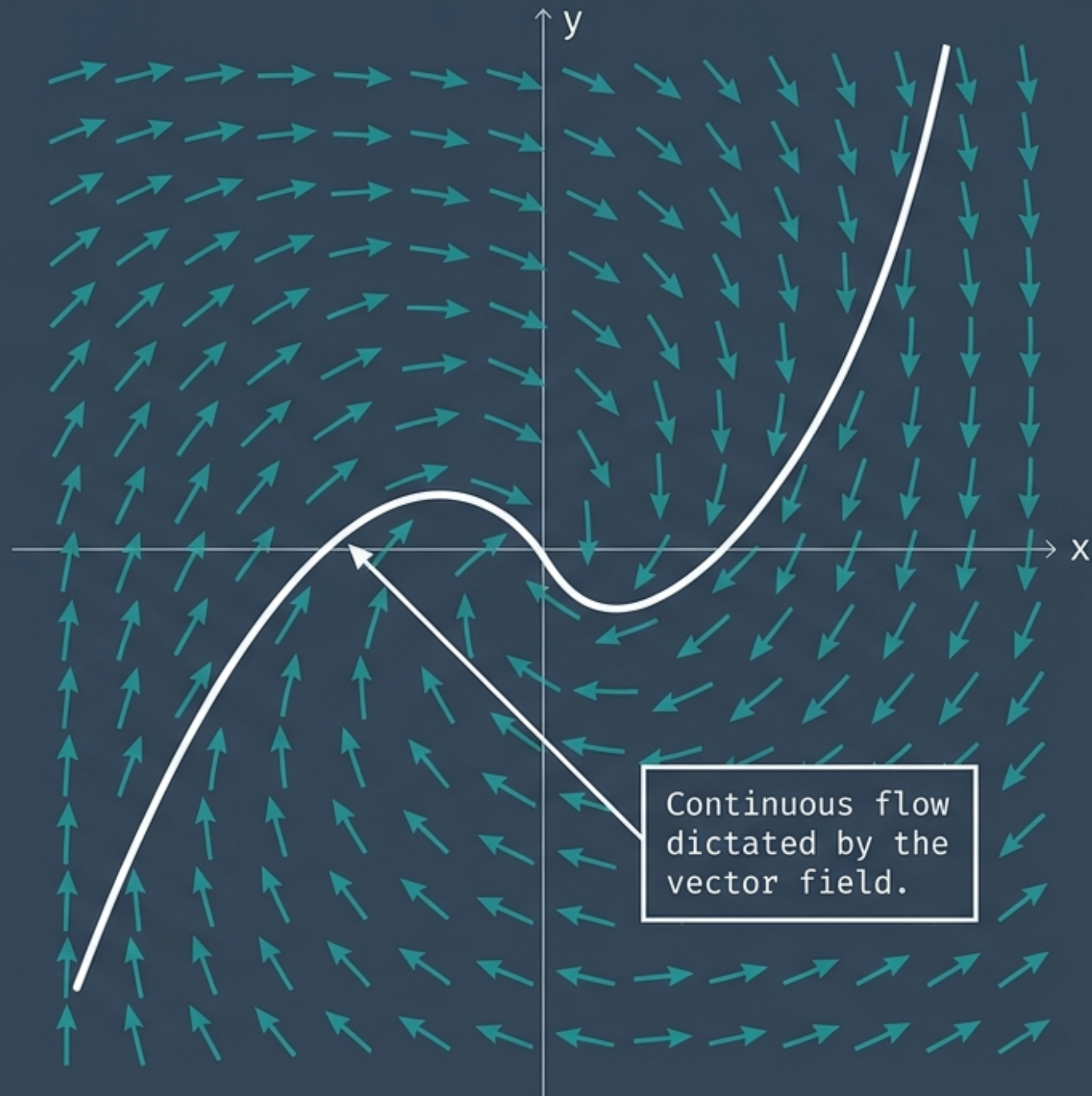


The System (Two Variables over Time)

$$\frac{dx}{dt} = P(x, y)$$

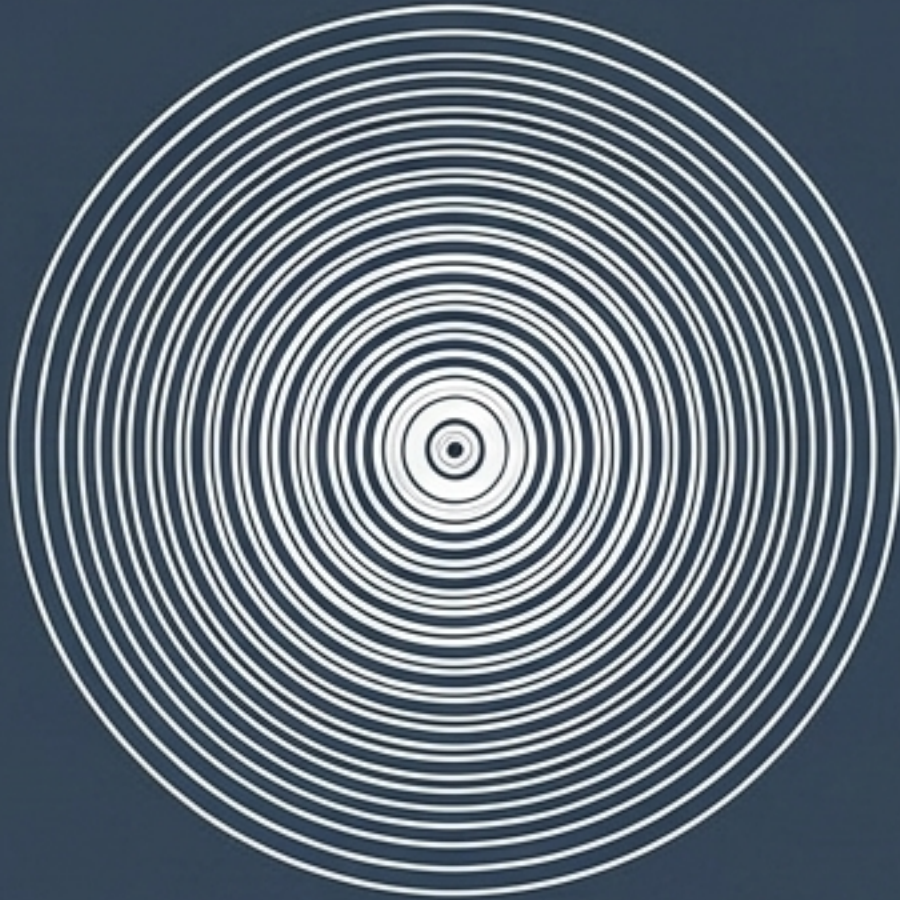
$$\frac{dy}{dt} = Q(x, y)$$

P and Q are real polynomials of degree d. At every point (x, y), the derivatives define a specific velocity vector.



What is a Limit Cycle? (Target: An isolated closed orbit)

The Non-Example: Center

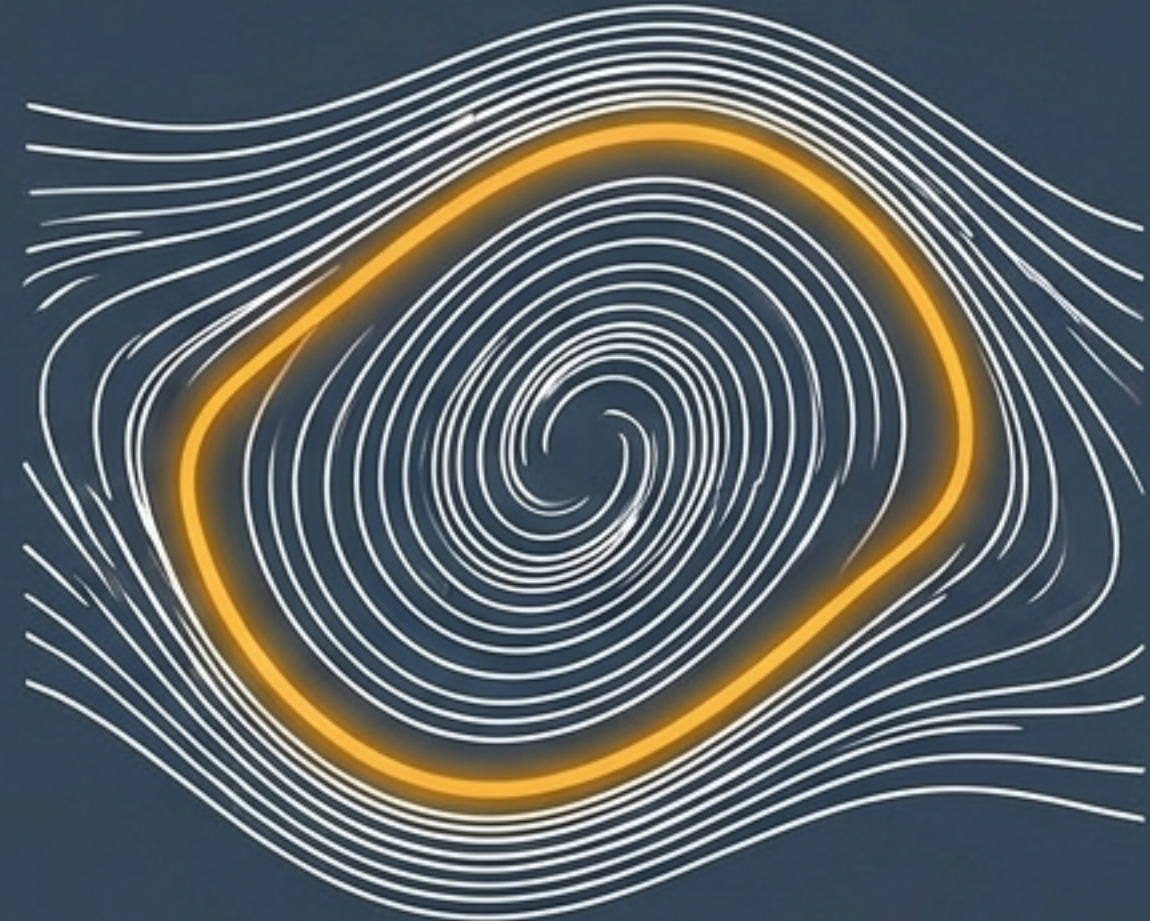


A continuum of periodic orbits.

NOT ISOLATED

Moving slightly off one orbit simply lands you on another parallel orbit.

The Example: Van der Pol

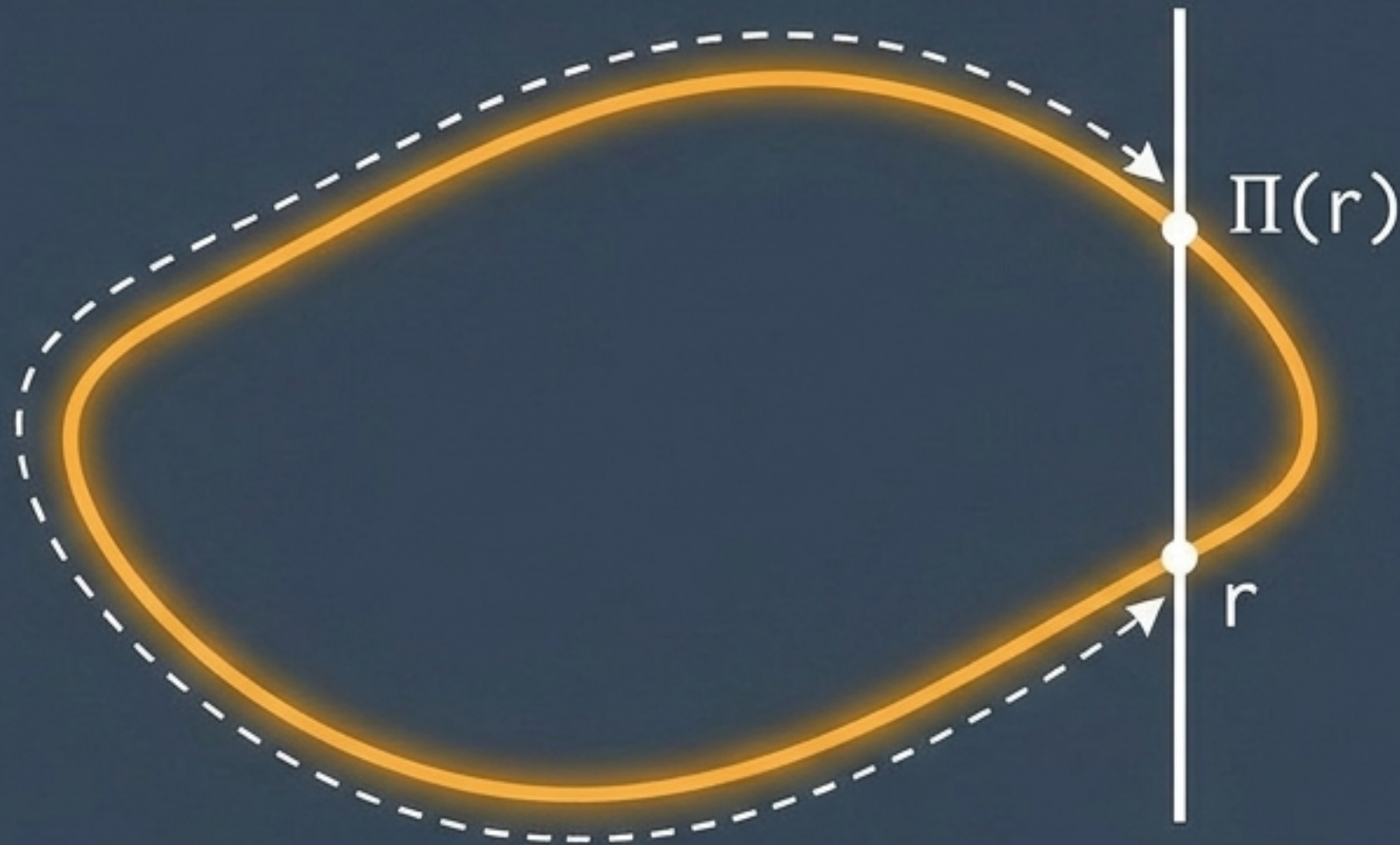


An isolated periodic solution.

ISOLATED (LIMIT CYCLE)

A slight perturbation forces the trajectory to ultimately return to this specific Amber cycle.

The Poincaré Section: Reducing the Dimension



MECHANICS: Return Map $\Pi(r)$:
Measures the position after
one full circuit from
starting point r .

Displacement Function:
 $\Pi(r) - r$

Displacement = 0

Translates to: Closed Orbits

Isolated Zeros

Translates to: Limit Cycles (Our Target)

The Century-Old Mystery: Hilbert's 16th Problem

1900: Paris International Congress of Mathematicians.

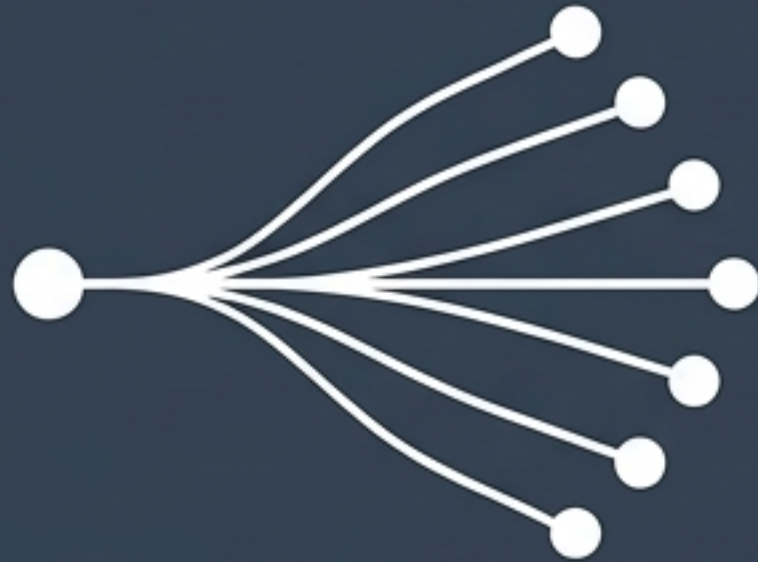
What is the maximum number and relative position of limit cycles for polynomial vector fields of degree n ?

$H(n)$

Does a finite upper bound exist depending only on the degree n ?
(Note: $H(1) = 0$ as linear fields have no limit cycles. For $n > 1$, the answer has remained unknown).

Individual Finiteness

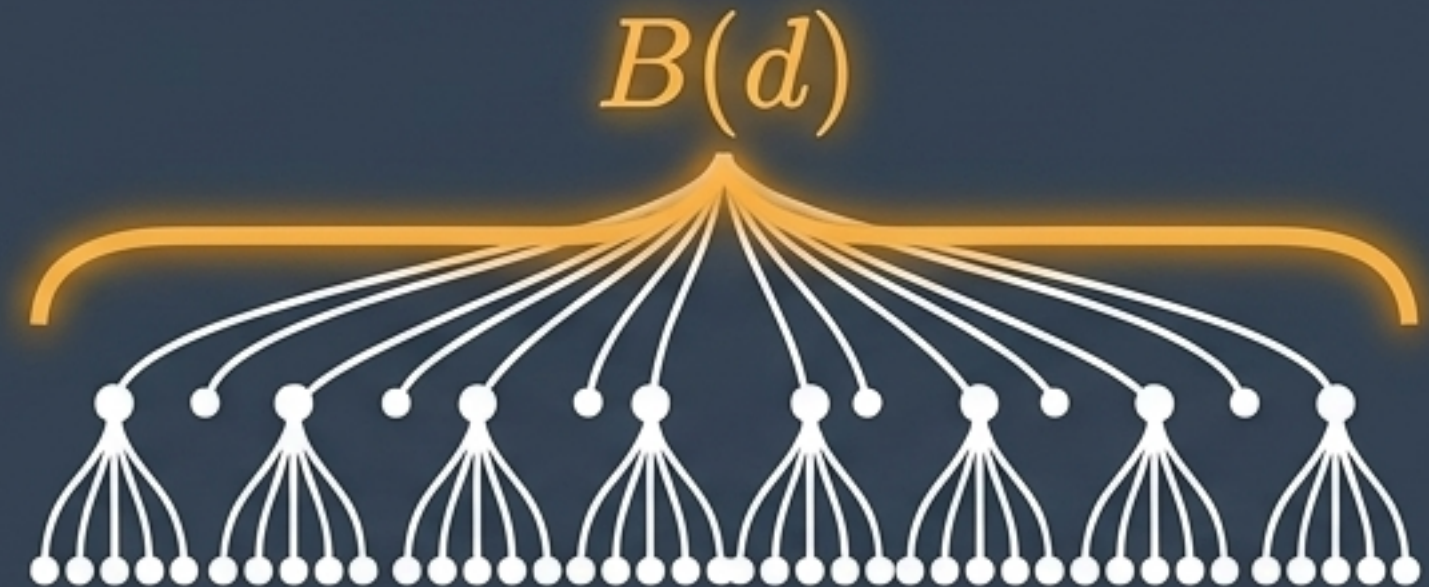
Does a single polynomial vector field have finitely many limit cycles?



- **1923:** Henri Dulac (Proof gap discovered in 1981).
- **1991 / 1992:** Yulii Ilyashenko & Jean Écalle (Independently proved: YES. Individual fields have finitely many cycles).

Uniform Boundedness

Is there a global cap $B(d)$ for every field of degree d ?



The Gap

Even if every individual field is finite, the maximum number across all fields of degree d could hypothetically be infinite.

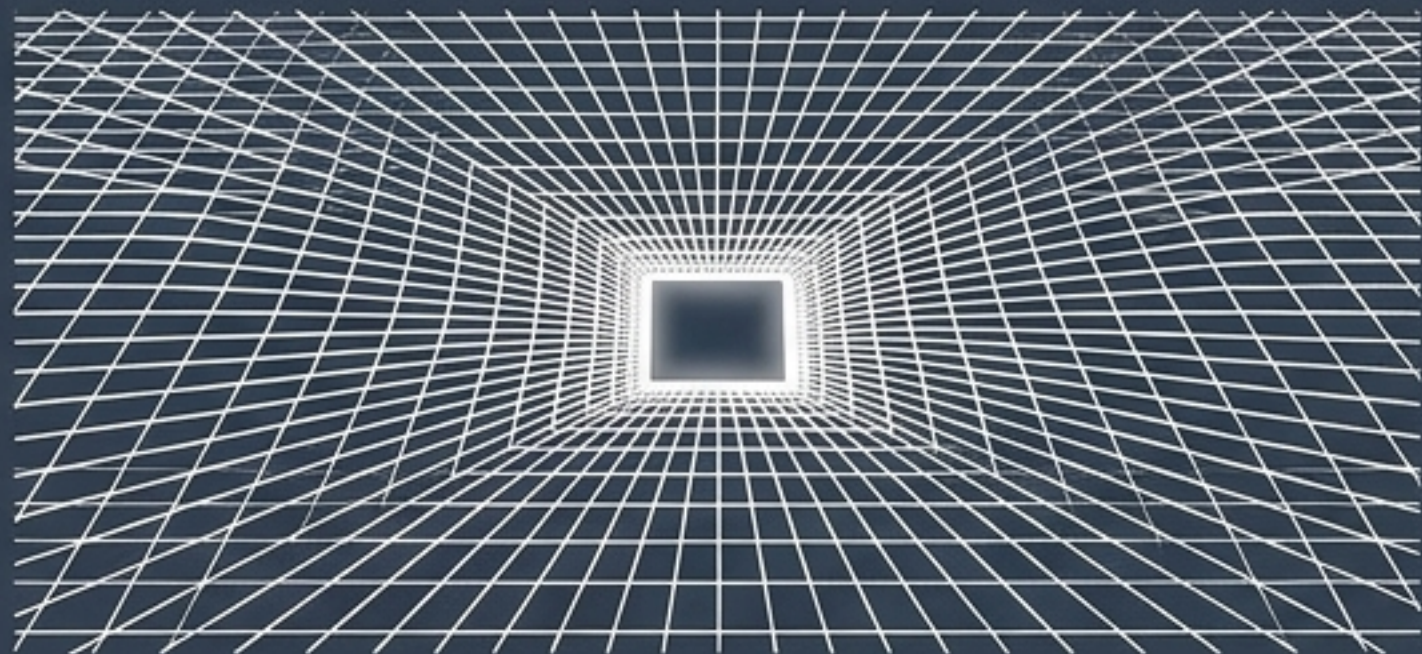
Status: Open until 2026.

“For each integer $d \geq 1$, there is a finite nonnegative integer $B(d)$ such that every real planar polynomial degree at most d has at most $B(d)$ limit cycles in the whole plane.”

RULE 1: This theorem proves a bound $B(d)$ EXISTS.

RULE 2: It gives NO VALUE for $B(d)$ or $H(n)$ —not even for $H(2)$.

Proof Step 1: Spatial Rescaling & Controlled Rotation



Rescaling

$$z = a + Ru$$

Bring all finite collections of limit cycles strictly inside one fixed bounded square.



Rotation

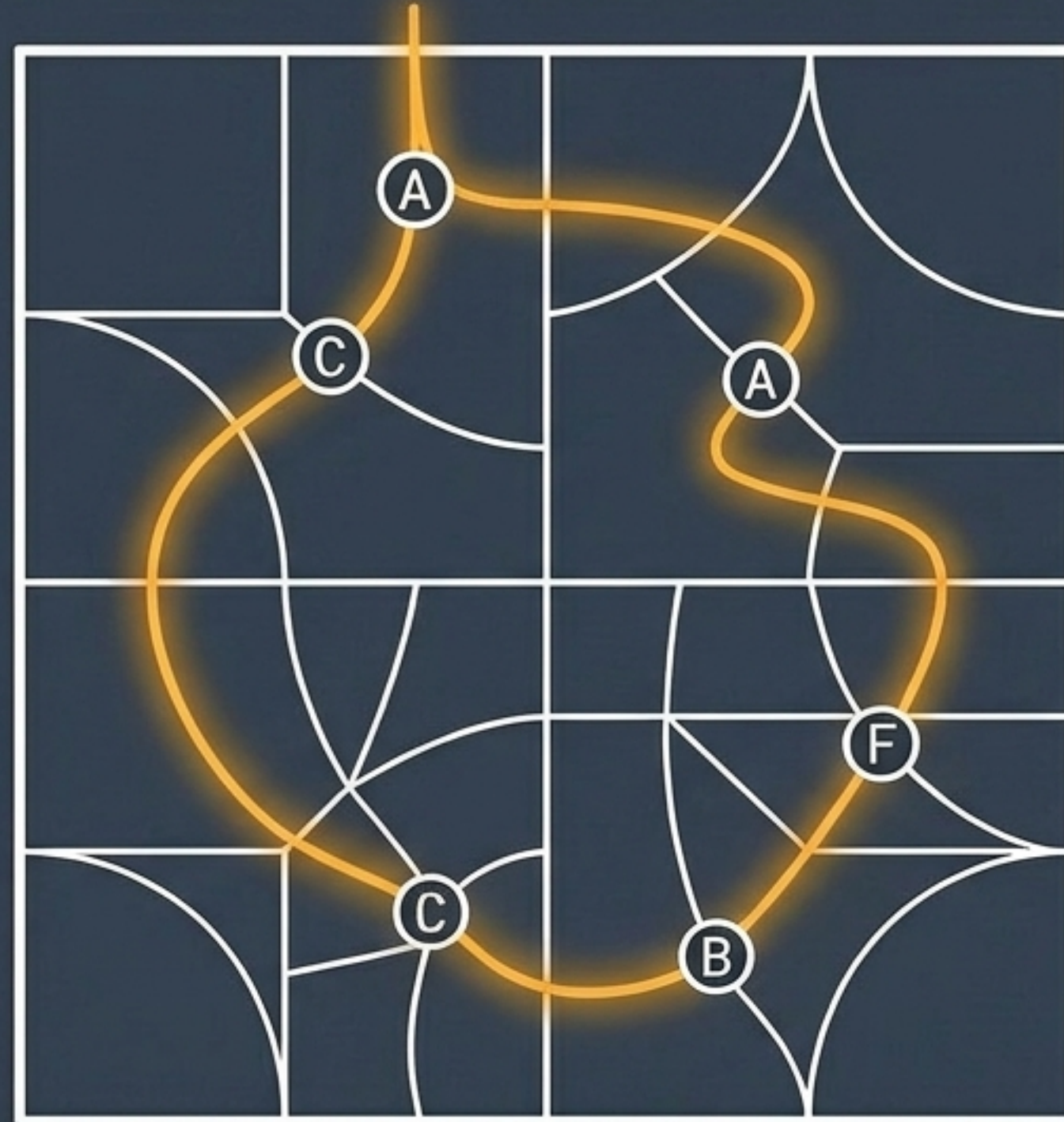
$$V_{\mu} = V + \mu(-Q, P)$$

A signed rotational perturbation guarantees the system produces at least as many hyperbolic (simple/isolated) cycles as any prescribed finite collection.

Proof Step 2: Finite Cell Types & Bounded Words

CELL DIVISION:

Cut the square into finitely many cell types (monomial pieces). Crucially, these shapes depend only on the degree d .

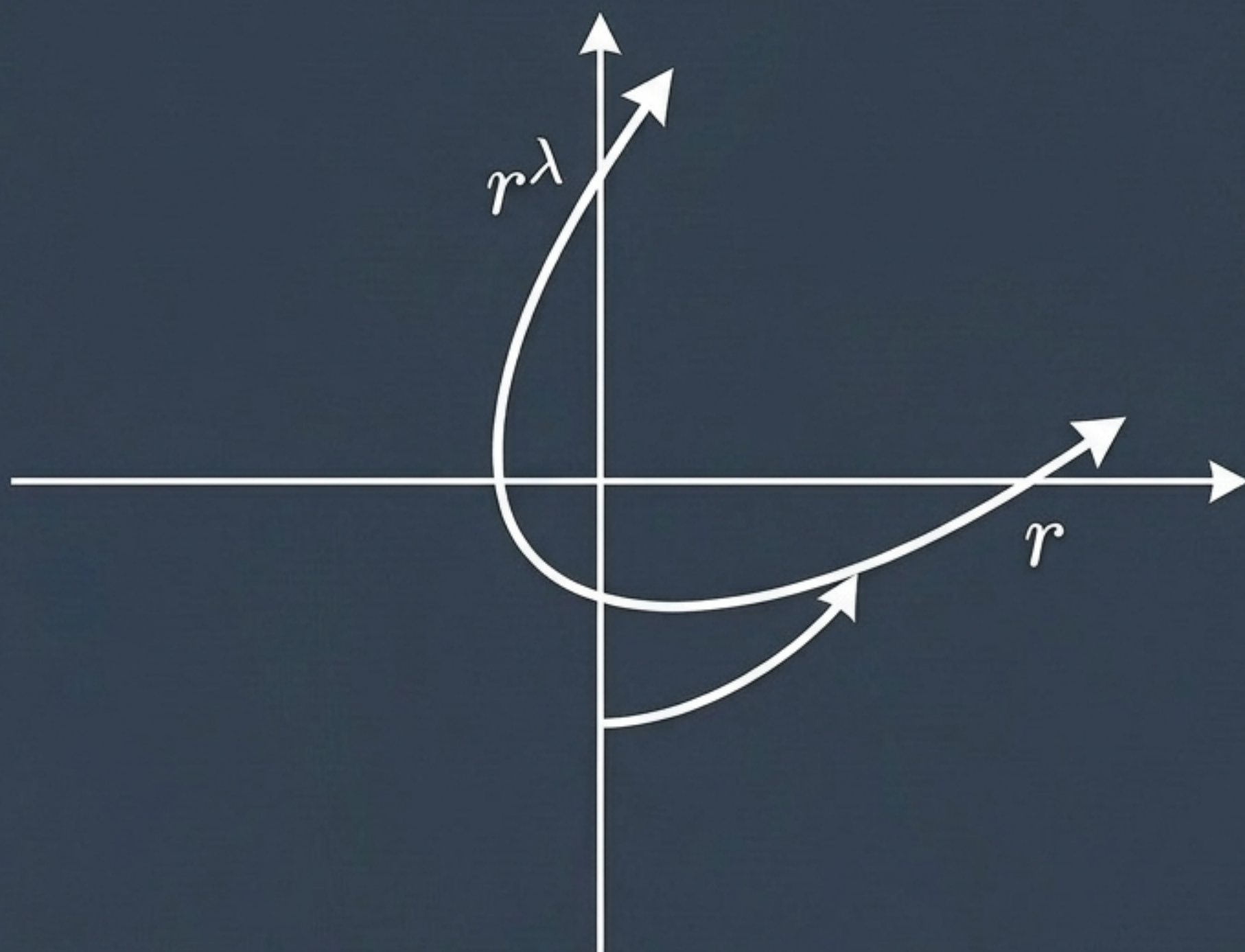


THE "WORD":

A limit cycle crosses each transverse boundary arc at most once.

Therefore, every continuous orbit is perfectly represented by a bounded, finite "word" of letters (e.g., $A \rightarrow C \rightarrow F \rightarrow B$).

Proof Step 3: Exact Scalar Transfers



THE LINK:

Each "letter" in the word represents exactly one of four scalar transfers.

EXAMPLE (SADDLE TRANSFER):

System: $x' = x, y' = -\lambda y$

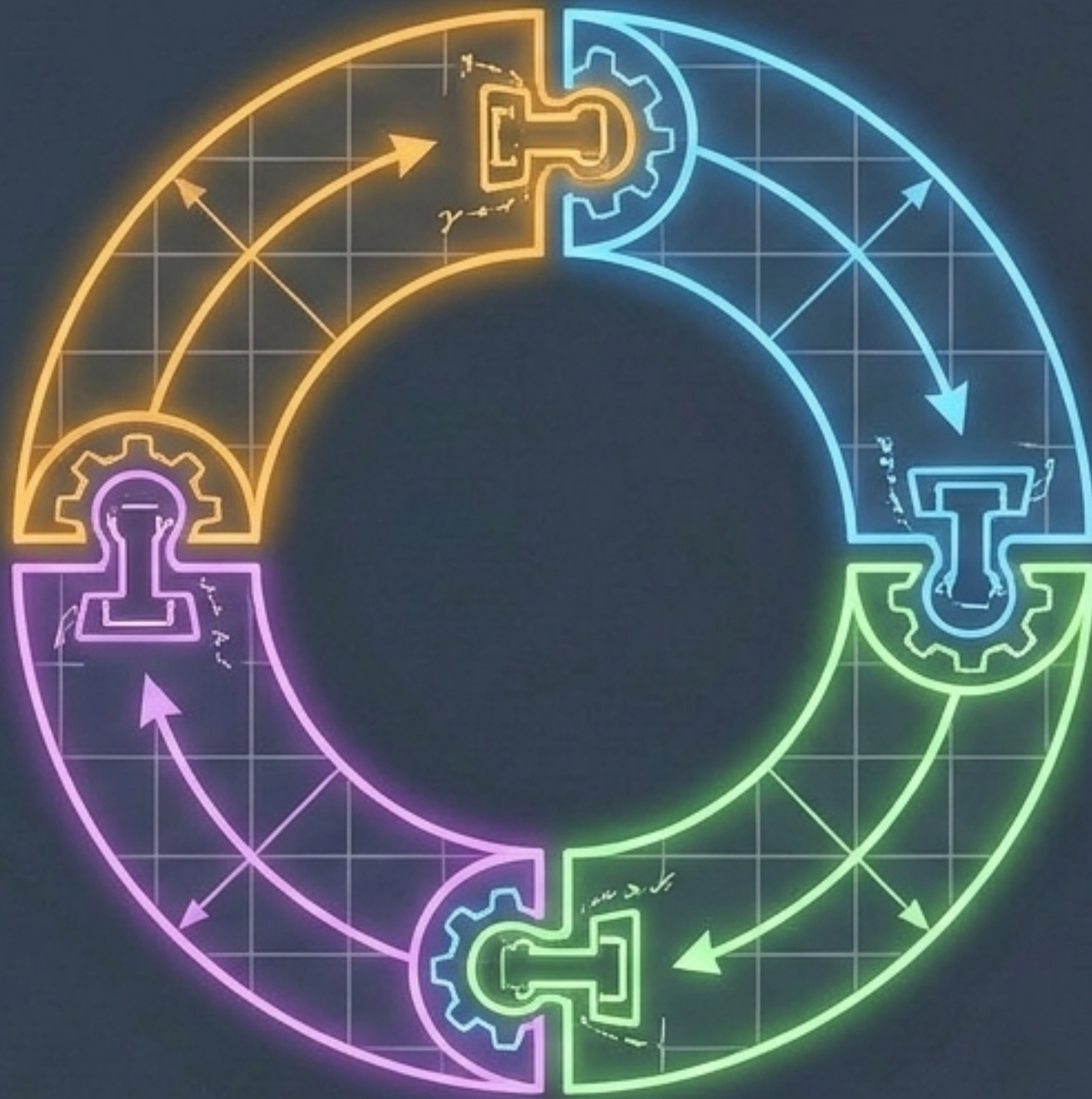
Mapping: $(r, 1) \rightarrow (1, r^\lambda)$

Flight Time: $\log(1/r)$

TAKEAWAY:

Intractable, dense geometry is reduced to precisely defined scalar equations.

Proof Step 4: Matching Equations & Separation



CLOSING THE LOOP:

For each valid "word", matching equations are generated to link the scalar transfers back together.

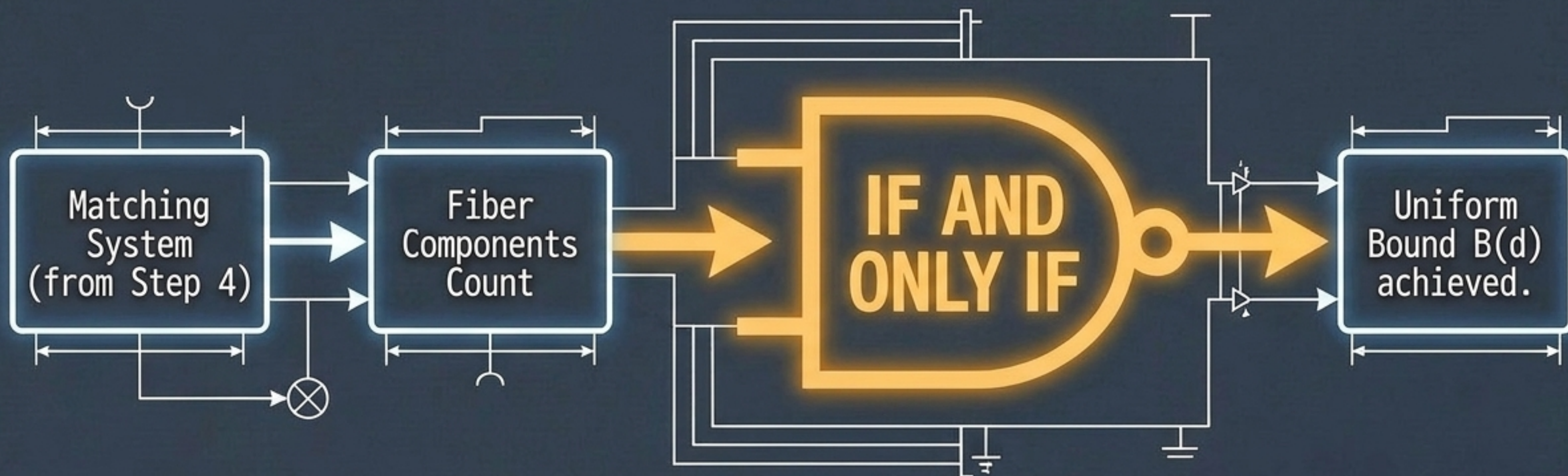
GEOMETRIC MEANING:

Exact local matching of these segments implies a closed periodic orbit.

THE SEPARATION PRINCIPLE:

Within each regular chart of these equations, representations of different hyperbolic limit cycles lie in different connected components of the fiber.

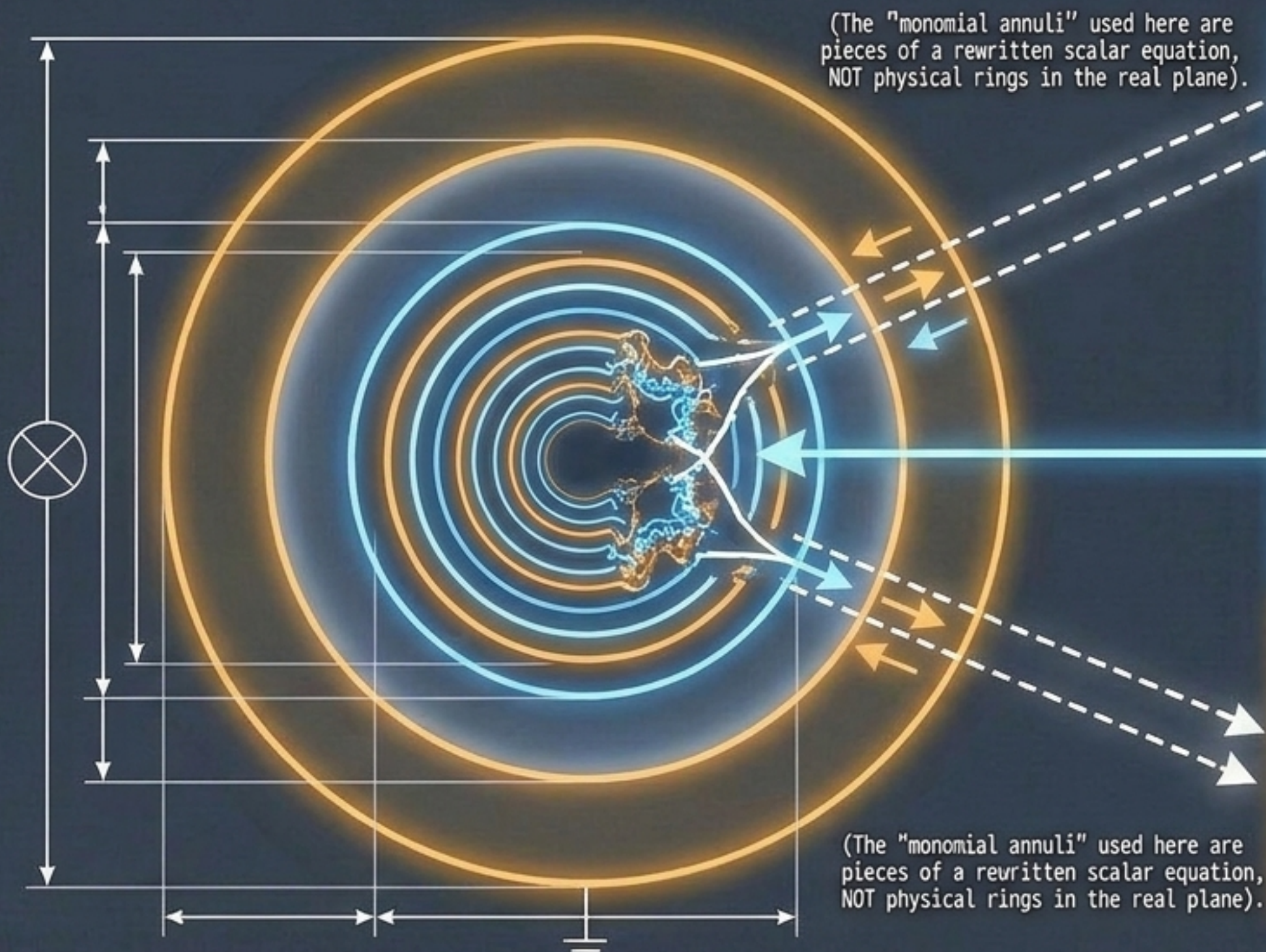
Proof Step 5: The Projection Count (The Fulcrum)



THE PRINCIPLE: Counting the connected components in the matching system uniformly bounds the number of limit cycles.

THE CATCH: This projection count only works if “absolute isolated-zero finiteness” holds for the closure of the generated systems. We must prove these isolated zeros are finite.

Proof Step 6: Asymptotic Packets & The Contradiction



THE SETUP:

Assume (for contradiction) an infinite sequence of isolated zeros exists.

THE MECHANISM:

Utilize exact "asymptotic packets" (finite saturated flags) evaluated on nested complex domains.

THE SEPARATION:

The calculus of these trees forces a separation, breaking the initial assumption. Finiteness is achieved.

The Computable Companion (Formal-Verified)

The Math

SYSTEM: Quintic Liénard

$$x' = y - F(x)$$

$$y' = -x$$

(where $\deg F \leq 5$)

THEOREM:

Yields at most 2 limit cycles.
The exact maximum is 2.

VERIFICATION:

Fully Lean-checked.

The Concrete Example

$$\text{FUNCTION: } F(x) = \varepsilon \left(4x - \frac{20}{3}x^3 + \frac{8}{5}x^5 \right)$$



Reality Check: What Is (and Isn't) Proved

WHAT WE HAVE

- A proof that a uniform finite bound $B(d)$ EXISTS for all degrees.
- A 160-page novel analytic proof utilizing asymptotic packets and matching equations.
- A Lean-checked proof that Quintic Liénard systems have exactly 2 limit cycles.

WHAT WE DON'T HAVE

- **NO VALUES:** NO exact value for $B(d)$ or $H(n)$ is provided (not even $H(2)$).
- **NO $H(5)$:** The Lean-verified '2' is not $H(5)$; it only applies to the restricted Liénard class.
- **NO FORMALIZATION:** The 160-page main proof is currently unformalized and unreviewed.
- **NO TRADITIONAL METHODS:** The main proof does not rely on Abelian integrals, Melnikov functions, computer searches, or Poincaré-Bendixson.